

# RAK5801 WisBlock 4-20mA Interface Module Datasheet

## Overview

### Description

The RAK5801 WisBlock Interface module was designed to be part of a production-ready IoT solution in a modular way and must be combined with a WisBlock Core and a Base module.

The RAK5801 is a 4-20 mA current loop extension module that allows the users to make an IoT solution for analog sensors with a 4-20 mA interface. This module converts the 4-20 mA current signal into voltage range supported by the WisBlock Core module (MCU) for further digitalization and data transmission.

The RAK5801 module features two input channels of 4-20 mA. Inside, a high-precision operational amplifier is used for signal amplification and conversion and supports a wide range of operating temperatures.

This module integrates a 12 V power supply, which can be used to power external sensors. The RAK5801 can be connected to 2-wire, 3-wire, or 4-wire types of 4-20 mA sensor. The module external interface is reached by a fast crimping terminal that allows connection for the 4-20 mA sensors (including power) and to the I2C bus. The fast crimping terminals can be used without the need of special tools, which simplifies the installation process on the field.

### Features

- Two 4-20 mA analog inputs
- Compatible with multiple WisBlock Core modules, such as RAK4631
- 0.1 mA conversion accuracy
- Supports low power consumption mode. The module can be powered off by the WisBlock Core module for saving energy during idle periods.

- 12 V output to power external sensors
- Reserved I2C expansion interface
- Fast crimping terminals
- Designed with a 2 kV ESD protection level
- Chipset: STMicroelectronics LM2902
- Small dimensions of 35 mm x 25 mm

## Specifications

### Overview

The overview discusses the block diagram of the board. It also shows the installation mechanism on how to mount the board into the baseboard.

### Block Diagram

The RAK5801 module was designed to convert 4-20 mA current signals into voltage signals by applying a sampling resistor. As shown in **Figure 1**, the input current signal from the sensor is conditioned by an operational amplifier to match the level supported by the ADC input of an MCU where the signal is digitized.

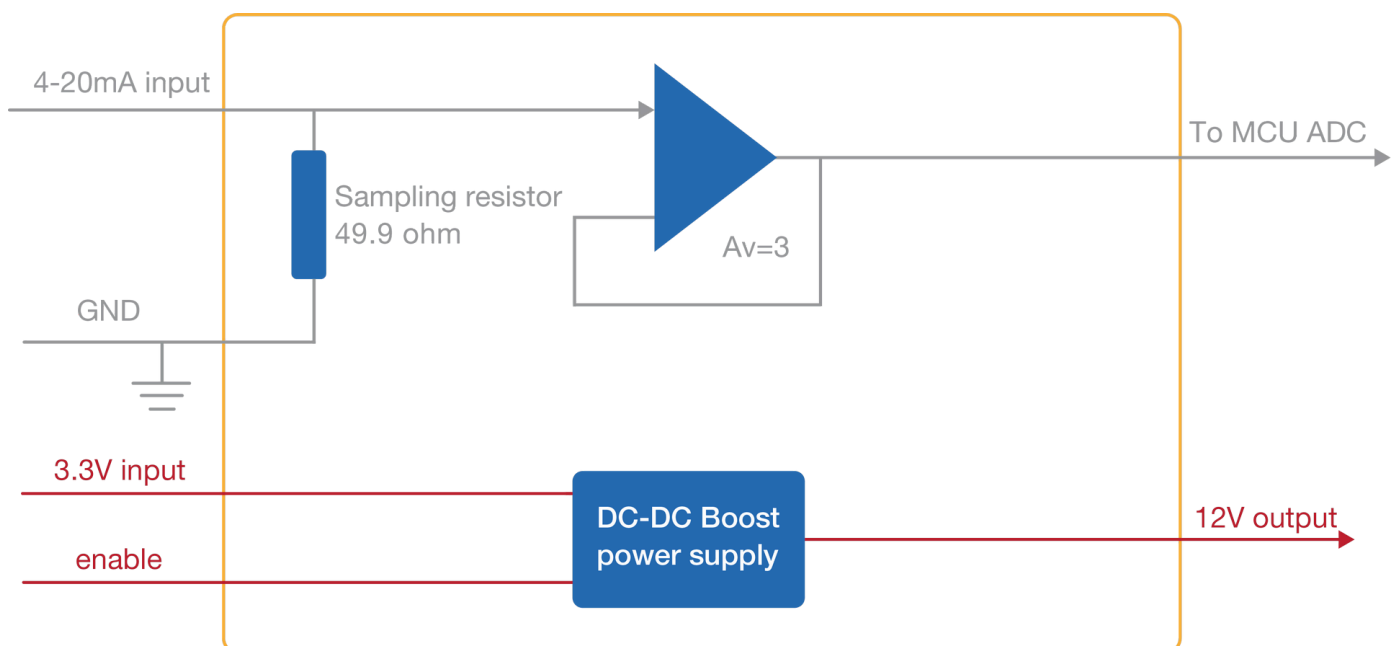


Figure 1: RAK5801 Block Diagram

Once the signal is digitalized, you can recover the original current value by applying the following formula:

$$I = U/149$$

Where **U** is the ADC reading and **I** the sensor current.

As shown in **Figure 1**, the module provides an output of 12 V for powering passive 4-20 mA sensors. This 12 V output is boosted by an internal DC-DC booster. The enable pin allows to control the power conversion module and set the RAK5801 module into a low power consumption mode.

## Hardware

The hardware specification is categorized into four parts. It discusses the pinouts of the board and its functionalities and diagrams.

## Chipset

| Vendor             | Part number |
|--------------------|-------------|
| STMicroelectronics | LM2902      |

## Device Specification

The following table shows the parameters and the description of the RAK5801 WisBlock 4-20 mA Interface Module:

| Parameter                  | Description           |
|----------------------------|-----------------------|
| Analog Input Interface     | 2 channels of 4-20 mA |
| Analog Sampling Resolution | 0.005 mA              |
| Analog Sampling Accuracy   | 1%                    |

| Parameter                        | Description                                                        |
|----------------------------------|--------------------------------------------------------------------|
| Analog Maximum Input Current     | 25 mA (There is a risk to burn the circuit surpassing this limit.) |
| Analog Port ESD Protection Level | 2 kV HBM                                                           |
| Current Sampling Resistor        | 49.9 $\Omega$                                                      |
| Operational Amplifier Gain       | 3.0                                                                |
| Input Voltage                    | 3.0-3.6 V                                                          |
| Output Voltage                   | 12 V                                                               |
| Output Current                   | Maximum 30 mA                                                      |
| Operating Temperature            | -30 °C ~ 65 °C                                                     |
| Storage Temperature              | -40 °C ~ 85 °C                                                     |
| Module Dimensions                | 35×25 mm                                                           |

## Pin Definition

This section covers the pin number of the sensor connector, the definition, and the functionalities of each pin shown in a tabular representation.

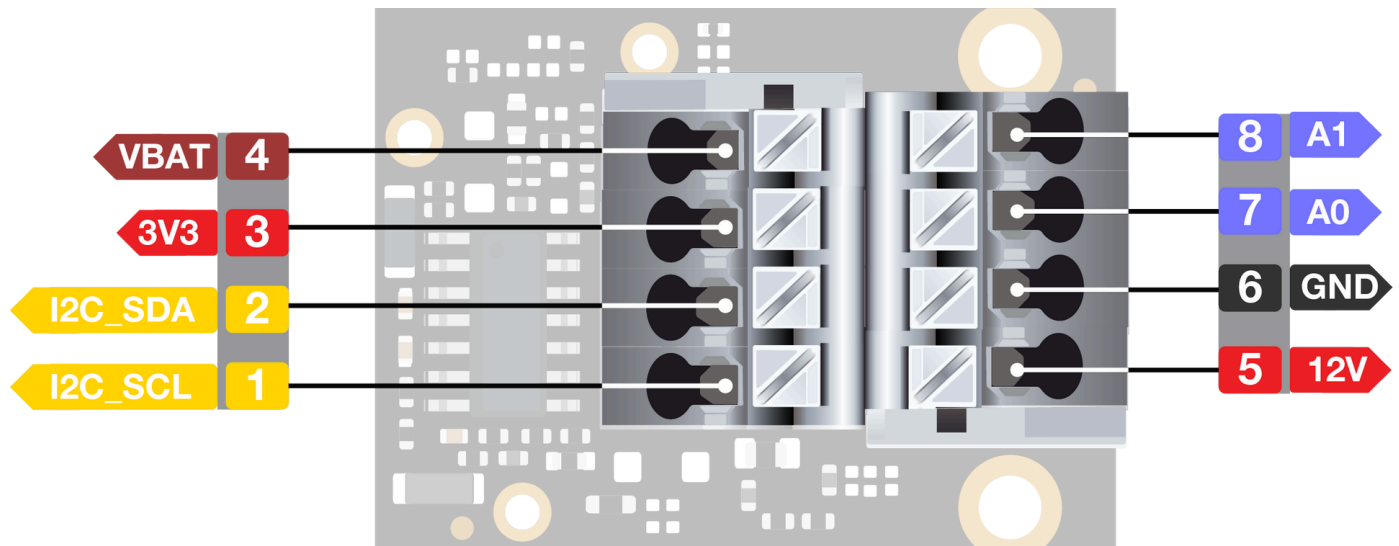


Figure 2: RAK5801 Sensor Connector

### NOTE

**A0** cannot be used as an analog input channel by default because it is used for measuring battery voltage. But if you need to use **A0**, there are few hardware modifications needed to configure.

To enable **A0** as an additional channel:

1. Remove **R7** on the WisBlock Base such as the RAK5005-O to disconnect Vbat sensing.
2. On RAK5801, remove the 0 Ohm resistor in **R94** and put it to **R95**. See **Figure 3**.

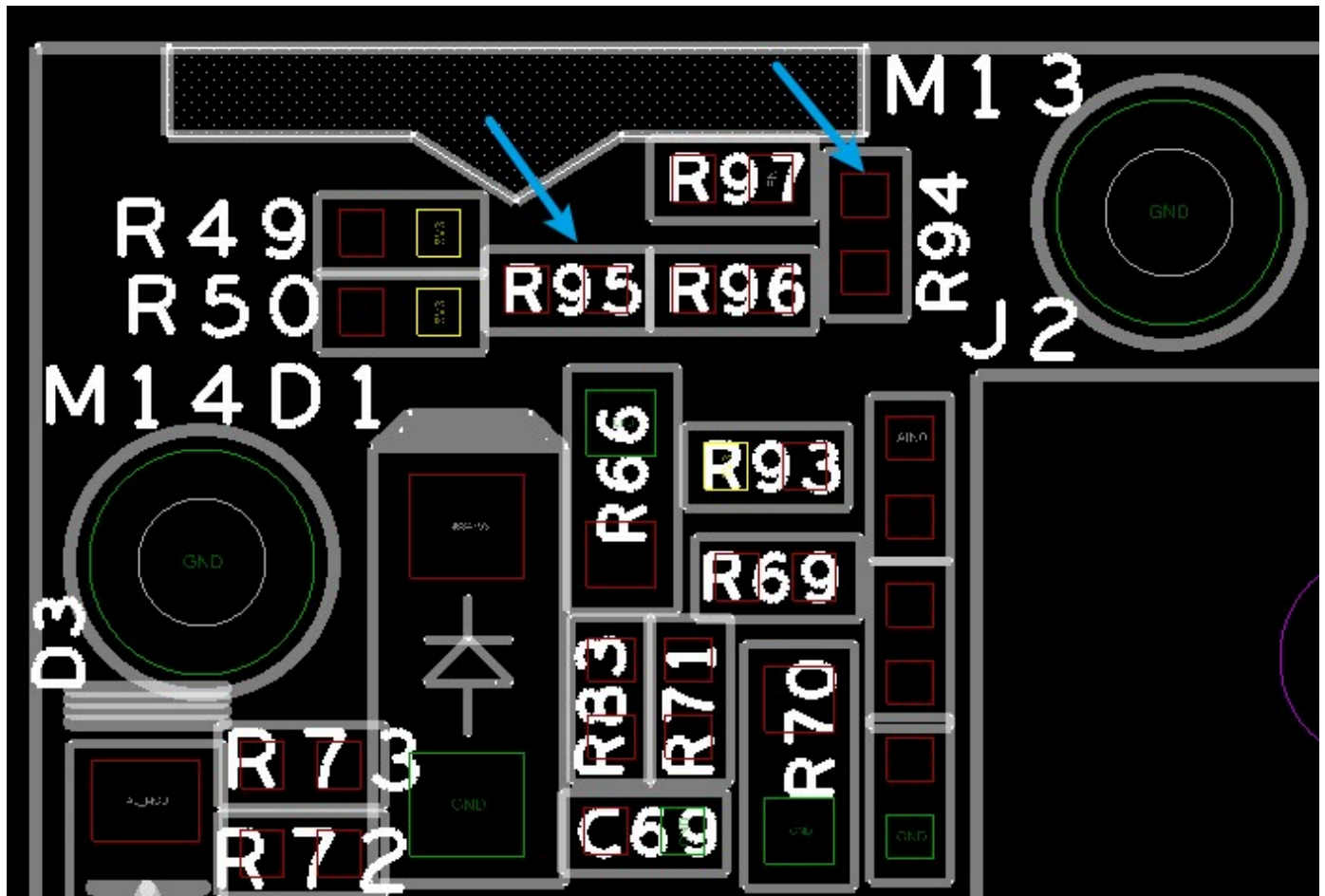
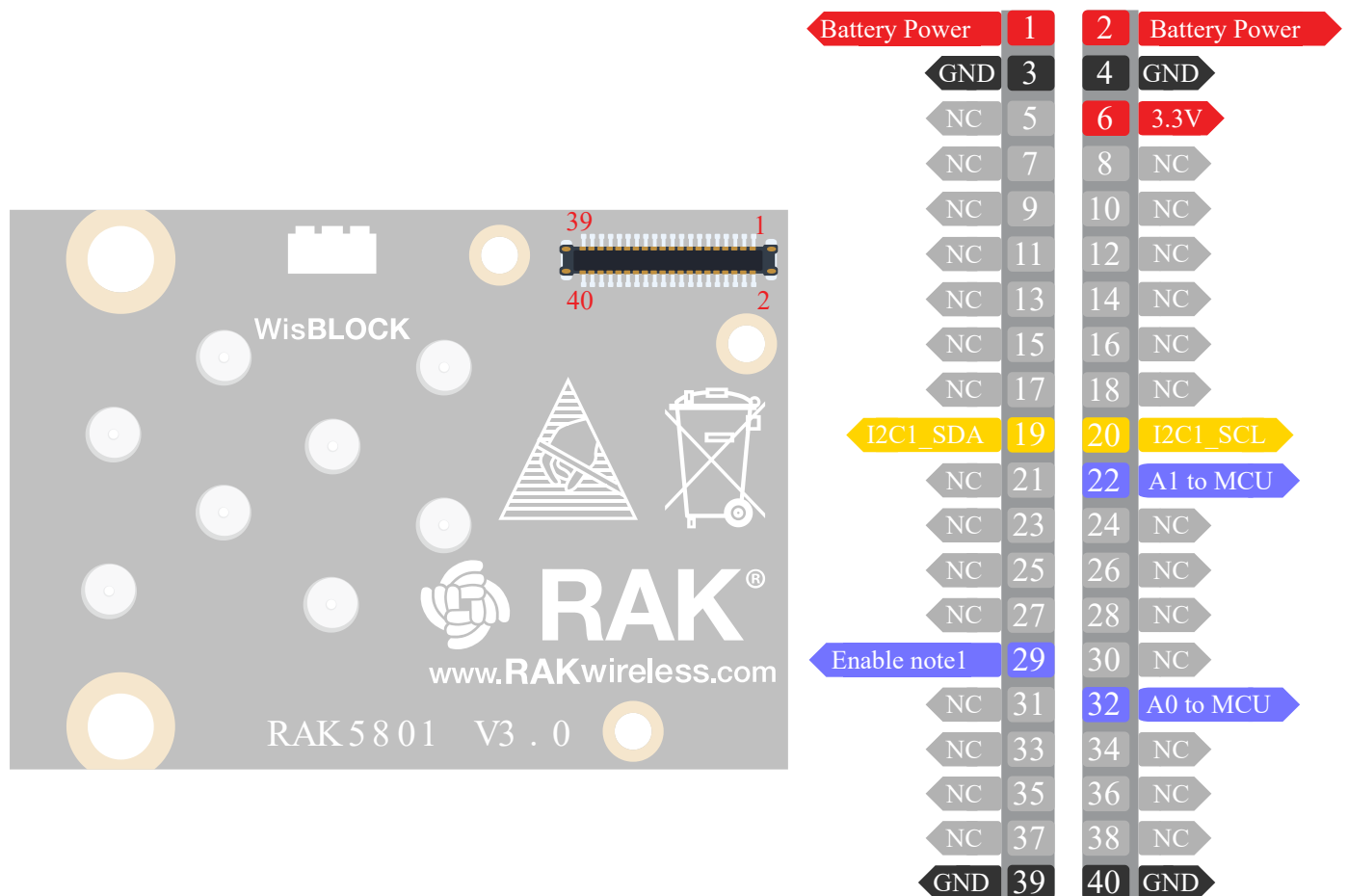


Figure 3: A0 Hardware Modifications

| Pin Number | Function Description             |
|------------|----------------------------------|
| 1          | SCL of the I2C interface         |
| 2          | SDA of the I2C interface         |
| 3          | 3V3 output                       |
| 4          | VBAT, Battery output             |
| 5          | 12 V output for external sensors |
| 6          | GND                              |
| 7          | Analog input 0                   |

| Pin Number | Function Description |
|------------|----------------------|
| 8          | Analog input 1       |

**Figure 4** shows the pin order for the IO connector of the module. Through this connector, the RAK5801 module is attached to the WisBoard baseboard.



**Figure 4:** RAK5801 Internal WisIO Connector

The functionalities of each pin of the WisIO connector are tabulated below:

| Pin Number | Description   | Pin Number | Description   |
|------------|---------------|------------|---------------|
| 1          | Battery Power | 2          | Battery Power |
| 3          | GND           | 4          | GND           |

| Pin Number | Description          | Pin Number | Description    |
|------------|----------------------|------------|----------------|
| 5          | NC, reserved for 3V3 | 6          | 3.3V Power     |
| 7          | NC                   | 8          | NC             |
| 9          | NC                   | 10         | NC             |
| 11         | NC                   | 12         | NC             |
| 13         | NC                   | 14         | NC             |
| 15         | NC                   | 16         | NC             |
| 17         | NC                   | 18         | NC             |
| 19         | SDA for I2C1         | 20         | SCL for I2C1   |
| 21         | NC                   | 22         | Analog to MCU  |
| 23         | NC                   | 24         | NC             |
| 25         | NC                   | 26         | NC             |
| 27         | NC                   | 28         | NC             |
| 29         | Enable note1         | 30         | NC             |
| 31         | NC                   | 32         | Analog0 to MCU |
| 33         | NC                   | 34         | NC             |
| 35         | NC                   | 36         | NC             |



| Pin Number | Description | Pin Number | Description |
|------------|-------------|------------|-------------|
| 37         | NC          | 38         | NC          |
| 39         | GND         | 40         | GND         |

### NOTE

This signal controls the dc-dc power supply on RAK5801, before capturing the analog signal, set this pin to high to enable power for RAK5801.

## Mechanical Characteristics

### Board Dimensions

Refer to **Figure 5** below for the mechanical dimensions of the RAK5801 module.

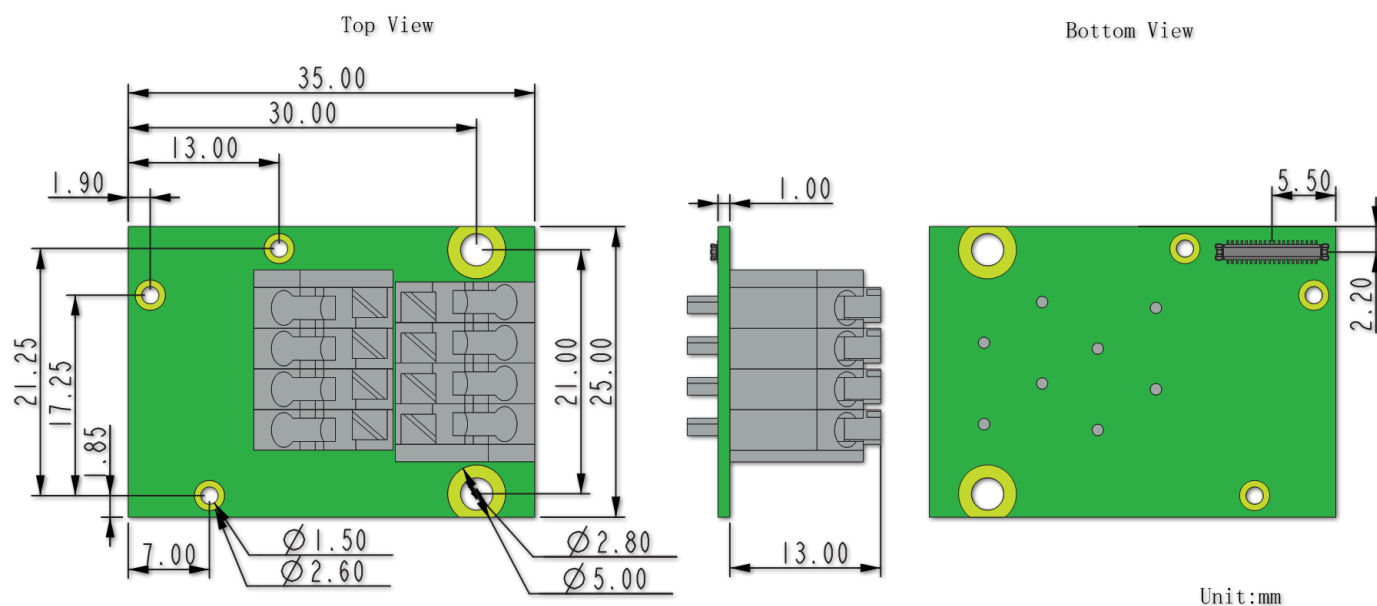


Figure 5: RAK5801 Mechanical Dimensions

## WisConnector PCB Layout

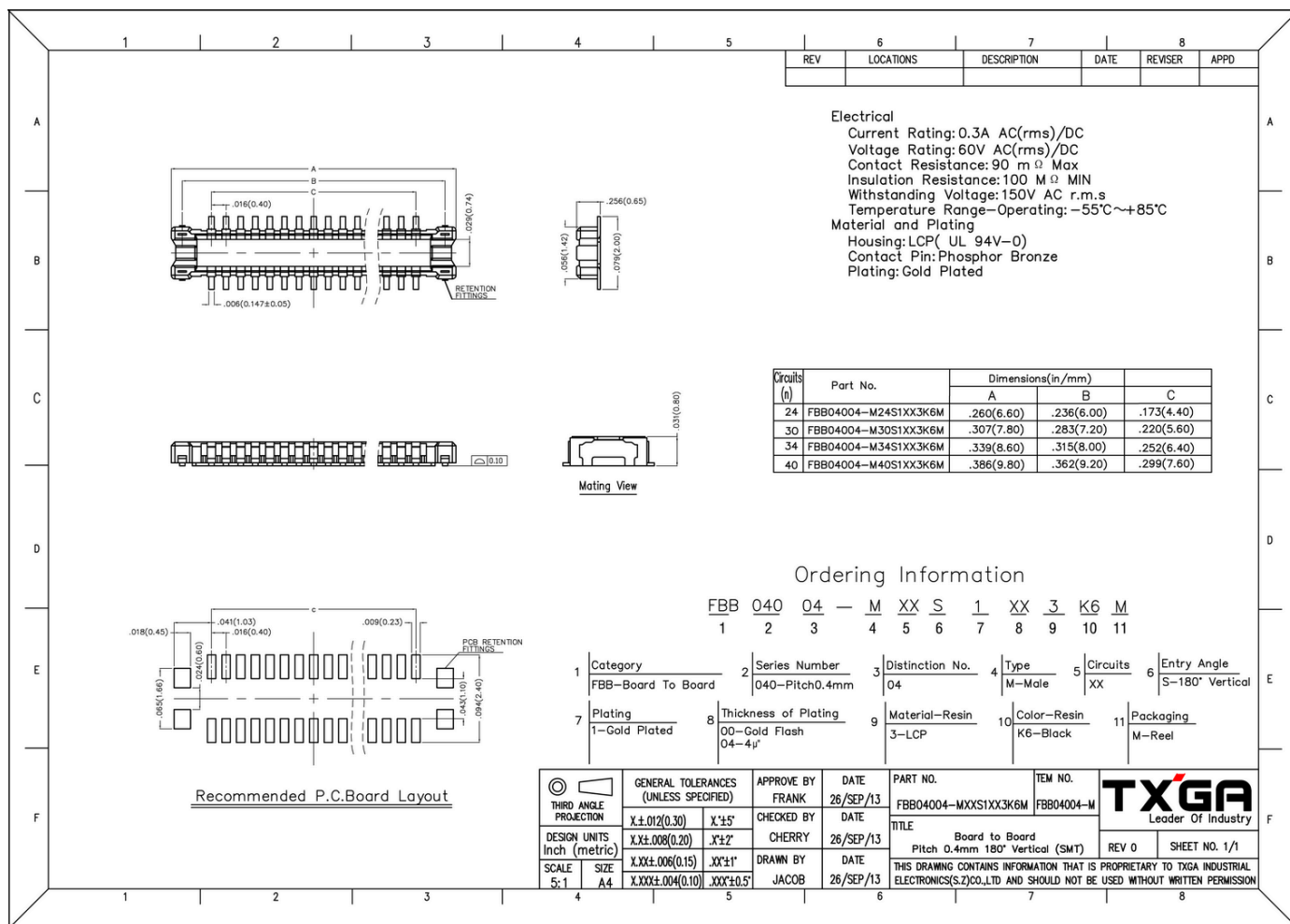


Figure 6: WisConnector PCB footprint and recommendations

## Schematic Diagram

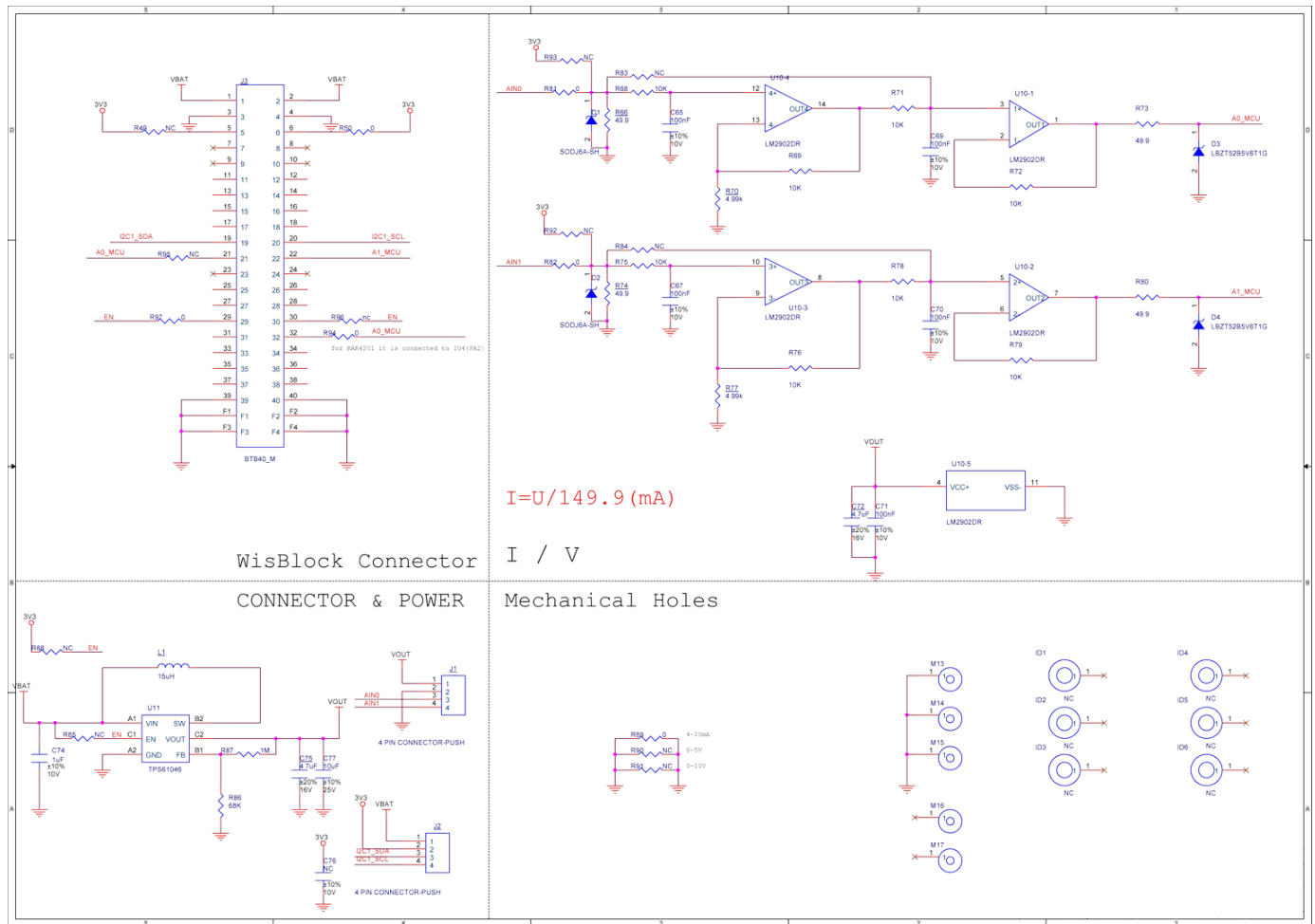


Figure 7: RAK5801 Schematic Diagram

## WisBlock Compatibility

Since a WisBlock module can be combined with a variety of different functional modules, the pin functions of the MCU are multiplexed, so the interface expansion module for each specific function may need to be properly adapted for the WisBlock. The compatibility details of the RAK5801 module are as shown in the table below:

| WisBlock Module    | Adaptable Module  | Description                                                                    |
|--------------------|-------------------|--------------------------------------------------------------------------------|
| WisBase Base board | RAK5005/RAK5005-O | RAK5801 is designed to be assembled in the IO slot of the RAK5005-O baseboard. |

| WisBlock Module      | Adaptable Module | Description                                                             |
|----------------------|------------------|-------------------------------------------------------------------------|
| WisBlock Core Module | RAK4631          | RAK5801 is compatible with RAK4631.                                     |
|                      | RAK4201          | Select RAK4201L-ADC for the low band or RAK4201H-ADC for the high band. |
|                      | RAK4202          | Refer to Note 2 for hardware adaptations to the RAK5005-O and RAK5801.  |
|                      | RAK4261          | Refer to Note 3 for hardware adaptations to the RAK5005-O and RAK5801.  |

#### NOTE

1. **The RAK5801+RAK4601** The RAK5801 is not compatible with RAK4601. The main reason is that RAK4601 doesn't expose any ADC pin through the RAK5005-O baseboard.

#### NOTE

##### 2. **RAK5801+RAK4202+RAK5005-O**

In order to combine a RAK5801 module, a RAK4202 (WisBlock Core module), and the RAK5005-O, the following modification must be introduced:

- In RAK5005-O, remove the R7 resistor, as shown in **Figure 8**.
- In RAK5801, remove R94 to R95 resistors. Use **PA0 of STM32L151** to read the analog data of the channel "analog0" and **PA2 of STM32L151** to read the analog data of Channel analog1. **Figure 9** shows the resistors R94 and R95 on the RAK5801 module.

This combination has the following restriction:

- The adapted RAK5005-O module will not be able to sense the battery voltage anymore.

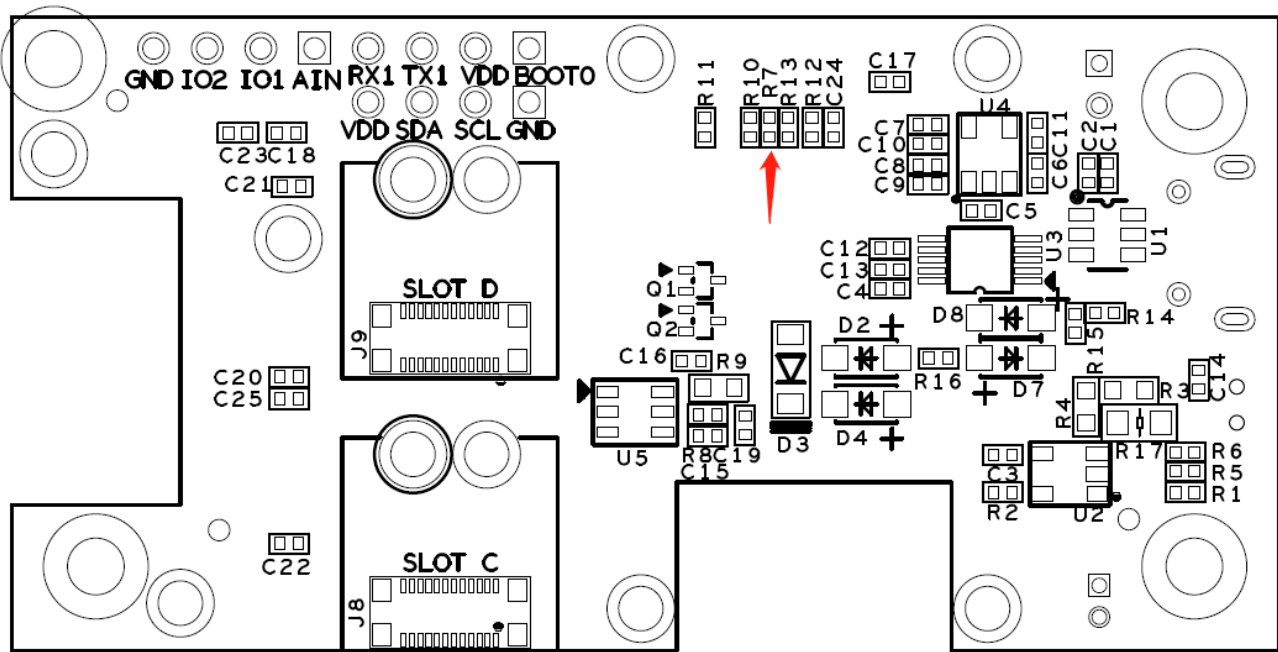


Figure 8: R7 on RAK5005-O

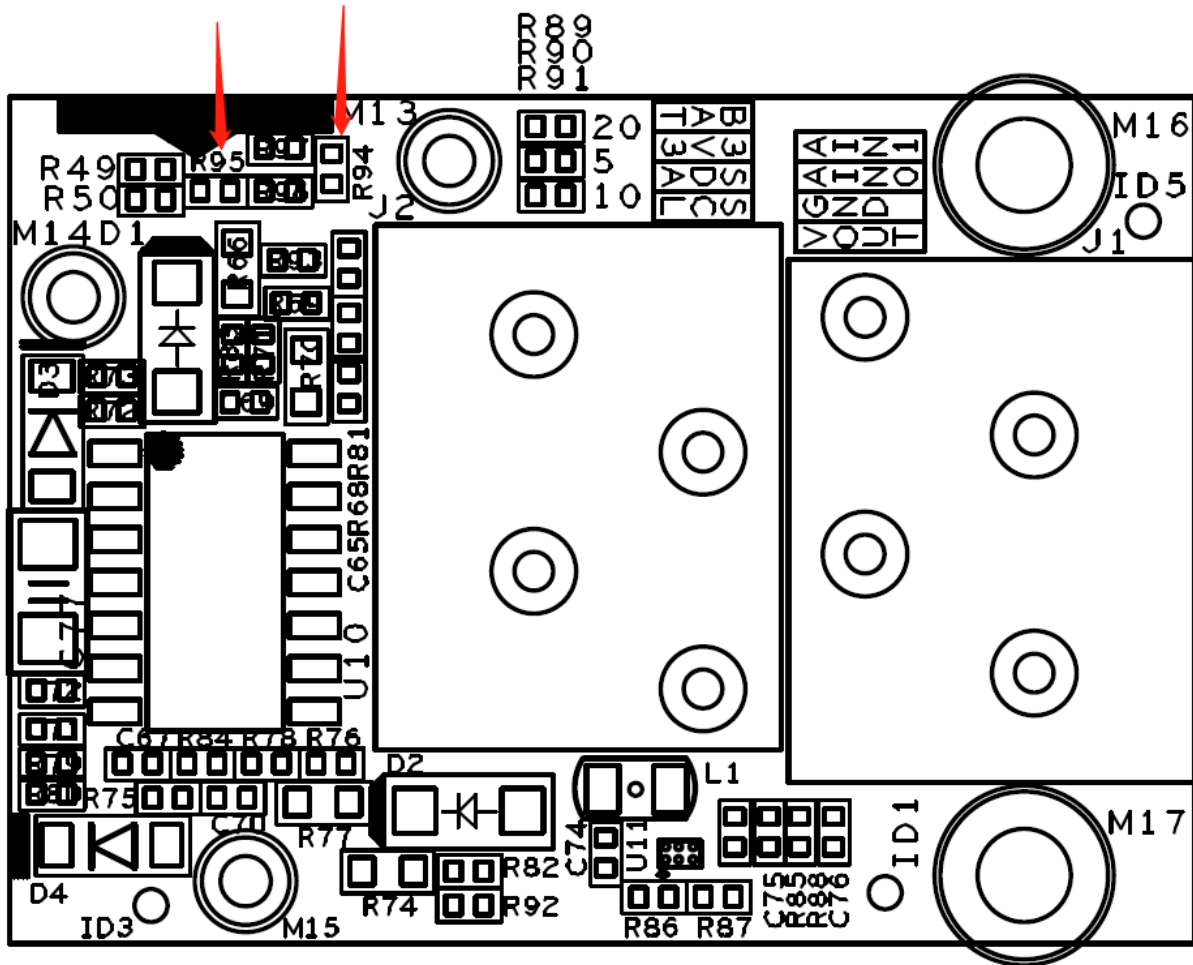


Figure 9: R94 and R95 on RAK5801

### NOTE

#### 3. RAK5801+RAK4261+RAK5005-O

In order to combine a RAK5801 module, a RAK4261(WisBlock Core module), and the RAK5005-O, the following modification must be introduced:

- In RAK5005-O, remove the R7 resistor. See **Figure 8**.
- In RAK5801, remove R94 to R95 resistors (see **Figure 9**). Use **PA08 of ATSAMR34** to read the analog data of the channel "analog0" and **PA09 of ATSAMR34** to read the analog data of Channel analog1.

This combination has the following restriction:

- The adapted RAK5005-O module will not be able to sense the battery voltage anymore.

